

EDUCATION	Department of Computer Science, Harvard University	Cambridge, MA
	<i>Ph.D. in Computer Science</i>	2020 - 2025
	• Advisors: Prof. Susan A. Murphy and Prof. Finale Doshi-Velez • GPA: 4.00/4.00. • Research area: reinforcement learning (RL) and Bayesian machine learning (ML)	
	Department of Computer Science, Harvard University	Cambridge, MA
	<i>M.S. in Computer Science</i>	2020 - 2024
	• GPA: 4.00/4.00. • Coursework: reinforcement learning, Bayesian machine learning, interpretable and explainable machine learning, probability theory	
	Department of Mathematics, University of California, Berkeley	Berkeley, CA
	<i>B.A. in Applied Mathematics</i>	2014 - 2018
	• GPA: 3.71/4.00. • Coursework: machine learning, artificial intelligence, algorithms & theory, linear algebra, numerical analysis, etc.	
EXPERIENCE	Applied Scientist II (RL & Recommendation Systems)	
	<i>Amazon Personalization</i>	2025 - Present
	Applied Scientist Intern (RL & Large Language Models)	
	<i>Amazon Alexa AI</i>	Summer 2023
	Research Scientist Intern (Data Science & Machine Learning)	
	<i>Etsy</i>	Summer 2021
	Technical Lead: Software Development Engineer	
	<i>Amazon</i>	2018 - 2020
SKILLS	Programming: Python, Pytorch, NumPy, Git, PyMC3, Java, Amazon Web Services (AWS), Xcode, React Native, Objective-C, JavaScript, Kotlin	
	Languages: English (native), Chinese (native), Spanish (proficient), Japanese (limited proficient)	
PATENTS		
	Miguel Roman Durazo Almeida, Thanhthanh Anh Tran, Jongun Jung, Rohin Patel, Anna L. Trella , Tarmandeep Toor, Shih-Ta Peng. <i>Item Identification based on receipt image</i> . U.S. Patent 11,257,049, filed January 31, 2020, and issued February 22, 2022.	
TEACHING		
	Graduate Teaching Fellow, Harvard University	
	<i>CS 181 Machine Learning (4.95/5.00 on student evaluations)</i>	2022
	Undergraduate Student Instructor, UC Berkeley	
	<i>CS 61C Machine Structures (4.91/5.00 on student evaluations)</i>	2017, 2018

AWARDS	• RLC Student Travel Fellowship, University of Edmonton, Alberta	2025
	• Global Symposium on AI in Dentistry Scholarship, Harvard University	2023
	• Derek Bok Certificate of Distinction in Teaching, Harvard University	2022
	• RLDM Student Travel Fellowship, Brown University	2022
	• EECS Outstanding Graduate Student Instructor Award, UC Berkeley	2018
	• Dean's Honors List, UC Berkeley	2016

INVITED TALKS	Oralytics: An Online Reinforcement Learning Algorithm Supporting Oral Self-Care <i>Harvard University, Guest Lecturer for STAT 234: Sequential Decision-Making</i>	2025
	A Deployed Online Reinforcement Learning Algorithm In A Clinical Trial <i>Conference on Statistical Learning and Data Science</i>	2024
	Monitoring Fidelity of Reinforcement Learning Algorithms in Clinical Trials <i>Society for Clinical Trials, Annual Meeting</i>	2024
	Oralytics: An Online Reinforcement Learning Algorithm Supporting Oral Self-Care <i>Harvard University, Guest Lecturer for STAT 234: Sequential Decision-Making</i>	2024
	Reward Design for Online Reinforcement Learning Algorithms <i>Two Sigma Headquarters, PhD Symposium</i>	2023
	Designing Reinforcement Learning Algorithms for Digital Interventions <i>University of Pennsylvania, Ivy Collective Engineering Symposium</i>	2022
	Online Reinforcement Learning Algorithms for Mobile Health <i>mHealthHUB, mDOT Center Webinar Series</i>	2022

- PUBLICATIONS**
1. **Anna L. Trella**, Walter Dempsey, Asim H. Gazi, Ziping Xu, Finale Doshi-Velez, Susan A. Murphy. “Non-Stationary Latent Auto-Regressive Bandits.” *Reinforcement Learning Journal (RLJ)*; selected for an oral at RLC, 2025.
 2. **Anna L. Trella**, Kelly W. Zhang, Hinal Jajal, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. “A Deployed Online Reinforcement Learning Algorithm In An Oral Health Clinical Trial.” *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI-25)*; selected for an oral, 2025.
 3. Susobhan Ghosh, Bhanu T. Gulapalli, Daiqi Gao, Asim H. Gazi, **Anna L. Trella**, Ziping Xu, Kelly Zhang, Susan A. Murphy. “Reproducible Workflow for Online AI in Digital Health.” *Philosophical Transactions of the Royal Society A*, 2025.
 4. **Anna L. Trella**, Ziming Li, Jinseok Nam, Puyang Xu. “Improving Personalization in Text Generation: A Two-Level Fine-Tuning Approach for LLMs.” *In Submission*.
 5. **Anna L. Trella**, Susobhan Ghosh, Erin E. Bonar, Lara Coughlin, Finale Doshi-Velez, Yongyi Guo, Pei-Yao Hung, Inbal Nahum-Shani, Vivek Shetty, Maureen Walton, Iris Yan, Kelly W. Zhang, Susan A. Murphy. “Effective Monitoring of Online Decision-Making Algorithms in Digital Intervention Implementation.” *In Submission*.
 6. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Iris Yan, Finale Doshi-Velez, Susan A. Murphy. “Monitoring Fidelity of Online Reinforcement Learning Algorithms in Clinical Trials.” *In Submission*.
 7. Kelly W. Zhang, Nowell Closser, **Anna L. Trella**, Susan A. Murphy. “Replicable Bandits for Digital Health Interventions.” *In Submission*.
 8. Esmail Keyvanshokoo, Kyra Gan, Yongyi Guo, Xueqing Liu, **Anna L. Trella**, Susan A. Murphy. “Improving Treatment Responses through Limited Nudges: A Data-Driven Learning and Optimization Approach.” *In Submission*.
 9. Inbal Nahum-Shani, Zara M. Greer, **Anna L. Trella**, Kelly W. Zhang, Stephanie M. Carpenter, Dennis Ruenger, David Elashoff, Susan A. Murphy, Vivek Shetty. “Optimizing an adaptive digital oral health intervention for promoting oral self-care behaviors: Micro-randomized trial protocol.” *Contemporary Clinical Trials*, 2024.
 10. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. “Reward Design For An Online Reinforcement Learning Algorithm Supporting Oral Self-Care.” *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI-23)*; selected for an oral, 2023.
 11. **Anna L. Trella**, Kelly W. Zhang, Inbal Nahum-Shani, Vivek Shetty, Finale Doshi-Velez, Susan A. Murphy. “Designing Reinforcement Learning Algorithms for Digital Interventions: Pre-Implementation Guidelines.” *Algorithms (Special Issue Algorithms in Decision Support Systems)*, 2022.
Preliminary version at *RLDM 2022 (Multi-disciplinary Conference on RL and Decision Making)*; selected for an oral (top 7% of accepted papers), 2022.
 12. **Anna L. Trella**, Peniel N. Argaw, Michelle M. Li, James A. Hay. “Discrepancies in Epidemiological Modeling of Aggregated Heterogeneous Data.” *IJCAI 2021 Workshop on Artificial Intelligence for Social Good*; selected for an oral, 2021.

ACADEMIC SERVICES

Reviewer: *Reinforcement Learning Conference (RLC) 2024*
Innovative Applications of Artificial Intelligence (IAAI) 2024
AAAI Conference on Artificial Intelligence (AAAI) 2026

HOBBIES

Figure Skating (Senior Moves in the Field Gold Medalist)
 Cheerleading (NCA National Champion, 2x California State Champion)
 Piano (Certificate of Merit Level 6)